Local Gauge Anomalies and The Standard Model

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Abstract

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2.1 Classical field theory

In classical mechanics we consider a countable set of particles each with finitely many degrees of freedom and generalised coordinates $q_i(t)$. These generalised coordinates $\{q_i(t)\}_i$ specify the system's configuration (position in configuration space) and together with the generalised conjugate momenta $\left\{p_i(t) \coloneqq \frac{\partial L}{\partial \dot{q}_i}\right\}_i$ define the system's position in phase space [1]. In classical field theory we generalise this notion of configuration space to a continuum with infinite degrees of freedom. The scalar field $\phi(t)$ can be seen as the generalised coordinates of a continuum $\left(q_i(t) \xrightarrow{i\to\vec{x}} \phi(\vec{x},t)\right)$ and for a system of continua the set of scalar fields $\{\phi_k(\vec{x},t)\}_k$ specifies the system's configuration [2]. Subsequently, we generalise the classical Lagrangian $L(q_i(t), \dot{q}_i(t), t)$ via

$$L(t) = \int d^3\vec{x}\, \mathcal{L}(\phi_k(\vec{x},t), \partial_\mu \phi_k(\vec{x},t), \vec{x}, t), \tag{2.1}$$

where $\mathcal{L}$ is the Lagrangian density. With respect to a path in configuration space, $\vec{\Phi}(\vec{x},t)$, the classical action becomes

$$S\left[\vec{\Phi}(\vec{x},t)\right] = \int dt\, L = \int d^4x\, \mathcal{L}(\phi_k(\vec{x},t), \partial_\mu \phi_k(\vec{x},t), \vec{x}, t). \tag{2.2}$$

Using Hamilton's principle [3]($\delta S[\vec{\Phi};\vec{\rho}] = 0, \forall \vec{\rho} \in C^\infty$) we obtain the field-theoretic Euler-Lagrange equations

$$\frac{\partial \mathcal{L}}{\partial \phi_k} \approx \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_k)}\right], \tag{2.3}$$

where we use $\approx$ to denote an on-shell equality. In Lagrangian systems the Hamiltonian, $H(p_i(t), q_i(t), t)$, is characterised by the Legendre transform

$$H = \sum_i p_i(t)\dot{q}_i(t) - L, \tag{2.4}$$

to generalise this to field theory we introduce the momentum field conjugate to $\phi_k(\vec{x},t)$

$$\pi_k(\vec{x},t) \coloneqq \frac{\partial \mathcal{L}}{\partial \dot{\phi}_k}, \tag{2.5}$$

$$p_i(t) \coloneqq \frac{\partial L}{\partial \dot{q}_i} \xrightarrow{i\to\vec{x}} \frac{\partial}{\partial \dot{\phi}_k}\left[\int d^3\vec{x}\, \mathcal{L}\right] = \int d^3\vec{x}\, \pi_k(\vec{x},t). \tag{2.6}$$

Thus the classical Hamiltonian is generalised to

$$H(t) = \int d^3\vec{x}\, \mathcal{H}(\pi_k(\vec{x},t), \phi_k(\vec{x},t), \vec{x}, t), \tag{2.7}$$

$$\mathcal{H} = \sum_k \pi_k(\vec{x}, t)\dot{\phi}_k(\vec{x}, t) - \mathcal{L}, \tag{2.8}$$

where $\mathcal{H}$ is the Hamiltonian density, and the field-theoretic analogue of Hamilton's equations [3] is the Hamiltonian field equations [2]

$$\dot{\phi}_k = \frac{\delta \mathcal{H}}{\delta \pi_k}, \quad \dot{\pi}_k = -\frac{\delta \mathcal{H}}{\delta \phi_k}, \tag{2.9}$$

where we use the functional derivative identify

$$\frac{\delta \mathcal{F}}{\delta g} = \frac{\partial \mathcal{F}}{\partial g} - \partial_\mu \left[\frac{\partial \mathcal{F}}{\partial(\partial_\mu g)}\right]. \tag{2.10}$$

To begin second quantisation we will also require the notion of a field-theoretic poisson bracket. Given two functionals, F and G, of the dynamical fields [1] given by

$$F = \int d^3\vec{x}\, \mathcal{F}(\pi_k, \phi_k, \vec{x}, t), \quad G = \int d^3\vec{x}\, \mathcal{G}(\pi_k, \phi_k, \vec{x}, t), \tag{2.11}$$

we define the poisson bracket in field theory via

$$\{F, G\}_f = \int d^3\vec{x} \sum_k \left[\frac{\delta \mathcal{F}}{\delta \phi_k}\frac{\delta \mathcal{G}}{\delta \pi_k} - \frac{\delta \mathcal{G}}{\delta \phi_k}\frac{\delta \mathcal{F}}{\delta \pi_k}\right] \tag{2.12}$$

2.2 Noether's first Theorem

In classical mechanics Noether's first theorem shows the correspondence between global symmetries of the lagrangian and conserved quantities called Noether charges. This generalises to global symmetries of the lagrangian density corresponding to conserved Noether currents in classical field theory. Consider an infinitesimal field transformation $\vartheta(\epsilon)$ such that

$$\phi_n \mapsto \phi_n + \delta_\epsilon \phi_n + \mathcal{O}(\epsilon^2), \quad \delta_\epsilon \phi_n = \epsilon \vartheta_n(\phi_k), \tag{2.13}$$

where the generators, ϑ_k, are independent of spacetime. A transformation is called a global symmetry if its effect on the Lagrangian density is

$$\mathcal{L} \mapsto \mathcal{L} + \epsilon \partial_\mu \Lambda^\mu(\phi_k, x) + \mathcal{O}(\epsilon^2). \tag{2.14}$$

This transformation changes the action by a surface term and the Euler-Lagrange equations are invariant. Using a Taylor expansion of the Lagrangian density under $\vartheta(\epsilon)$ we find that

$$\delta_\epsilon \mathcal{L} = \sum_k \delta_\epsilon \phi_k E_k + \partial_\mu \theta^\mu, \tag{2.15}$$

$$\theta^\mu = \sum_k \delta_\epsilon \phi_k \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_k)}. \tag{2.16}$$

[1] n.b. $K(x) = \int dy\,[K(y)\delta(x-y)]$

Where E_k are the Euler-Lagrange derivatives and θ^μ is the symplectic potential density [4]. By considering the shift in the action we arrive at Noether's first theorem [5, 6]; an infinitesimal transformation $\vartheta(\epsilon)$ is a symmetry of the Lagrangian if and only if

$$j^\mu(\epsilon) \coloneqq \theta^\mu - \epsilon\Lambda^\mu, \quad \partial_\mu j^\mu = -\delta_\epsilon\phi_k E_k \approx 0. \tag{2.17}$$

Where we use the summation convention for the field indices hereafter and the on-shell divergence free quantity j^μ is called the conserved Noether current. This easily generalises to local symmetries: spacetime dependent generators $\vartheta_k(\phi, x)$.

2.3 Noether's Second Theorem

Noether's second theorem shows that a local symmetry introduces off-shell constraints on the equations of motion and leads to Noether current identities which are crucial for gauge theories. Under local infinitesimal symmetries R_ξ of the form

$$\phi_n \mapsto \phi_n + \delta_\xi\phi_n + \mathcal{O}(\xi^2), \quad \delta_\xi\phi_n = \sum_{m=0} R_n^{\mu_1\cdots\mu_m}(\phi_k)\partial_{\mu_1}\cdots\partial_{\mu_m}\xi(x), \tag{2.18}$$

the action shifts by

$$\delta_\xi S = \int d^4x\, \delta_\xi\phi_n E_n + \partial_\mu\theta^\mu. \tag{2.19}$$

If $\xi(x)$ is a compactly supported smooth function, the boundary term vanishes and we have

$$\int d^4x\, \delta_\xi\phi_n E_n = 0, \tag{2.20}$$

and integration by parts leads to

$$\int d^4x\, \xi(x)\Delta(E) = 0, \quad \Delta(E) \coloneqq \sum_{m=0} (-1)^m\partial_{\mu_1}\cdots\partial_{\mu_m}[R_n^{\mu_1\cdots\mu_m}(\phi_k)E_n] \tag{2.21}$$

By the fundamental lemma of the calculus of variations [7] we obtain the off-shell constraint $\Delta(E) = 0$. Thus the equations of motion are not all independent and the Euler-Lagrange equations ($E_n = 0$) are under-determined. However, observables must be uniquely determined by the theory hence we conclude some degrees of freedom are gauge. Using the surface term S^μ from integration by parts we find an expression similar to eq2.17

$$\delta_\xi\phi_n E_n = \partial_\mu S^\mu(E_k, \xi), \tag{2.22}$$

with the important difference that S^μ vanishes on-shell. The quantity $j^\mu + S^\mu$ is divergence-free off-shell, subsequently by the Poincaré lemma we obtain Noether's second theorem [5, 6, 8, 9]; on-shell the canonical Noether current j^μ is the divergence of a two-form

$$j^\mu(\xi) = -S^\mu(E_k, \xi) + \partial_\nu\kappa^{\mu\nu}(\xi) \approx \partial_\nu\kappa^{\mu\nu}(\xi), \quad \kappa^{\mu\nu}(\xi) = -\kappa^{\nu\mu}(\xi) \tag{2.23}$$

the anti-symmetric two-form $\kappa^{\mu\nu}$ is called a superpotential. Although the superpotential is arbitrary ($\kappa^{\mu\nu}$ is determined up to addition by a divergence-free quantity) it can be uniquely determined if one demands the superpotential obeys additional constraints (such as asymptotic conservation) [9]. In the case of electrodynamics the superpotential is electromagnetic field tensor and eq2.23 becomes Maxwell's equations

$$J^\mu = \partial_\nu F^{\nu\mu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{2.24}$$

3 The Dirac Lagrangian

3.1 Gauging the vector symmetry

In quantum mechanics the Schrodinger equation is not Lorentz invariant and is thus incompatible with special relativity. A generalisation of the relativistic dispersion relation using Hamiltonian and momentum operators for free particles leads to the Klein-Gordon equation [10, 11]. However, the Klein-Gordon equation is a second order partial differential equation and subsequently does not uniquely determine time-evolution of the wavefunction. Alternatively, by finding the half-iterate of d'Alembert operator the Klein-Gordon equation simplifies to a first order partial differential equation; the Dirac equation [12]

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0. \tag{3.1}$$

We call the corresponding Lagrangian density the Dirac Lagrangian

$$\mathcal{L}_\mathrm{D} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi, \tag{3.2}$$

where the adjoint bispinor is defined as $\bar{\psi} \coloneqq \psi^\dagger \gamma^0$. This Lagrangian admits a global U(1) symmetry called the vector symmetry

$$\psi \mapsto e^{i\vartheta}\psi, \quad \bar{\psi} \mapsto \bar{\psi} e^{-i\vartheta}. \tag{3.3}$$

This symmetry corresponds to a conserved current, namely the vector current, via Noether's first theorem

$$j^\mu = \bar{\psi}\gamma^\mu \psi. \tag{3.4}$$

We now attempt to 'gauge' this symmetry by adding spacetime dependence to the infinitesimal parameter

$$\psi \mapsto e^{i\vartheta(x)}\psi, \quad \bar{\psi} \mapsto \bar{\psi} e^{-i\vartheta(x)}. \tag{3.5}$$

However, the Lagrangian is not invariant under such a transformation so we introduce a derivative operator which transforms covariantly

$$D_\mu \coloneqq \partial_\mu + ie\Pi_\mu, \tag{3.6}$$

$$D_\mu \psi \mapsto D'_\mu \left[e^{i\vartheta(x)}\psi \right] = e^{i\vartheta(x)} D_\mu \psi. \tag{3.7}$$

To satisfy this the gauge field must transform as

$$\Pi_\mu \mapsto \Pi_\mu - \frac{1}{e}\partial_\mu \vartheta(x), \tag{3.8}$$

Thus we have a modified Lagrangian which admits a local U(1) gauge symmetry

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\Pi_\mu j^\mu, \tag{3.9}$$

where $\not{D} := \gamma^\mu D_\mu$ is the Dirac operator. This is strikingly similar to the electromagnetic Lagrangian density which also has a U(1) gauge symmetry

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_\mu J^\mu. \tag{3.10}$$

Thus we identify $\Pi_\mu = A_\mu$ and $ej^\mu = J^\mu$ and arrive at the Lagrangian for quantum electrodynamics.

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - eA_\mu j^\mu + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi. \tag{3.11}$$

Schwinger arrived at the covariant formulation of quantum electrodynamics similarly by constructing a Lagrangian invariant under Lorentz transformations, gauge transformations, and charge conjugation [13]. However, this is not the end of the story, the massless Dirac equation admits another global U(1) symmetry. In the next sections we construct a gauge-invariant Lagrangian and investigate the resultant gauge theory.

3.2 Gauging the axial symmetry

The massless Dirac Lagrangian admits another symmetry better seen in the Weyl basis

$$\psi = \begin{pmatrix} \psi_{\text{L}} \\ \psi_{\text{R}} \end{pmatrix}, \quad \gamma^\mu\partial_\mu = \begin{pmatrix} 0 & \sigma^\mu\partial_\mu \\ \bar{\sigma}^\mu\partial_\mu & 0 \end{pmatrix}. \tag{3.12}$$

In this basis the Dirac Lagrangian can be seen as the interaction between two Weyl fermions of opposite chirality [14]

$$\mathcal{L}_{\text{D}} = \psi_{\text{L}}^\dagger(i\sigma^\mu\partial_\mu)\psi_{\text{L}} + \psi_{\text{R}}^\dagger(i\bar{\sigma}^\mu\partial_\mu)\psi_{\text{R}} - m(\psi_{\text{L}}^\dagger\psi_{\text{R}} + \psi_{\text{R}}^\dagger\psi_{\text{L}}). \tag{3.13}$$

In the massless case the Lagrangian gains a global U(1) symmetry known as the axial symmetry

$$\psi_{\text{L}} \mapsto e^{i\vartheta}\psi_{\text{L}}, \quad \psi_{\text{R}} \mapsto e^{-i\vartheta}\psi_{\text{R}}, \tag{3.14}$$

which is equivalent to

$$\psi \mapsto e^{i\vartheta\gamma^5}\psi, \tag{3.15}$$

and a Noether current

$$j_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi, \quad \partial_\mu j_5^\mu = 2im\bar{\psi}\gamma^5\psi. \tag{3.16}$$

We now proceed à la Schwinger, gauging the axial symmetry by promoting ∂_μ to a covariant derivative $\widetilde{D}_\mu$ as in eq3.6 and determining the required gauge transformation

$$\mathcal{L}_D|_{m=0} = \bar{\psi}(i\gamma^\mu\partial_\mu)\psi - q\bar{\psi}\gamma^\mu\Pi_\mu\psi, \tag{3.17}$$

$$\Pi_\mu \mapsto e^{i\vartheta\gamma^5}\Pi_\mu e^{-i\vartheta\gamma^5} - \frac{1}{q}\partial_\mu\vartheta\gamma^5. \tag{3.18}$$

Now we must add a kinetic term to the Lagrangian to determine the dynamics of the matrix-valued gauge field. Such a term must be gauge-invariant to preserve the U(1) symmetry. We find a Lagrangian analogous to the Maxwell Lagrangian

$$\mathcal{L}_\Pi = -\frac{1}{4}\mathrm{Tr}(\Gamma_{\mu\nu}\Gamma^{\mu\nu}), \tag{3.19}$$

with field strength

$$\Gamma_{\mu\nu} = \partial_\mu\Pi_\nu - \partial_\nu\Pi_\mu + iq[\Pi_\mu, \Pi_\nu], \tag{3.20}$$

has the required gauge-invariance and is Lorentz invariant. Thus we have constructed a classical theory describing the interaction between massless dirac fermions and an axial gauge boson

$$\mathcal{L}_\mathrm{A} = -\frac{1}{4}\mathrm{Tr}(\Gamma_{\mu\nu}\Gamma^{\mu\nu}) + \bar{\psi}i\widetilde{\not{D}}\psi, \tag{3.21}$$

$$\mathcal{L}_\mathrm{A} = -\frac{1}{4}\mathrm{Tr}(\Gamma_{\mu\nu}\Gamma^{\mu\nu}) - q\bar{\psi}\gamma^\mu\Pi_\mu\psi + \bar{\psi}(i\gamma^\mu\partial_\mu)\psi. \tag{3.22}$$

4 Quantum Field Theory

So far we have studied relativistic quantum mechanics, making use of classical fields to describe the dynamics of quantum states. However, such a quantum theory is inconsistent and predicts negative energy states which require peculiar interpretations to reconcile like Dirac's infamous hole theory [15]. To proceed to a quantum theory of particle interactions we must promote the dynamical fields to field operators; second quantisation. We shall see how this shift to a quantised field theory affects our calculation of expectation values and probabilities.

4.1 Second Quantisation

In classical mechanics, the canonical transformations are exactly those which preserve the symplectic structure; invariance of poisson brackets of the dynamical variables (p_i, q_i) characterises transformations which leave Hamilton's equations unchanged [3]. A system's position in phase space specifies its classical state. However, in quantum mechanics all properties of a system are included in a quantum state, $|\psi\rangle$, inside of a Hilbert space upon which operators corresponding to observables act.

Dirac's famous canonical quantisation rule ($\{A, B\} \to \frac{1}{i\hbar}[\hat{A}, \hat{B}]$) [12] allows us quantise the canonical structure of classical mechanics (although Groenewold's theorem shows us that such a rule cannot hold for all functions of the dynamical variables [16]). To obtain a quantised field theory we consider the field-theoretic poisson brackets

$$\{\phi_n(z), \phi_m(w)\}_f = 0 = \{\pi_n(z), \pi_m(w)\}_f \tag{4.1}$$

$$\{\phi_n(z), \pi_m(w)\}_f = \delta_{nm}\delta^{(3)}(\vec{z} - \vec{w}) \tag{4.2}$$

Which are quantised to the canonical commutation relations for the dynamical field operators acting on a Fock space

$$[\hat{\phi}_n(z), \hat{\phi}_m(w)] = 0 = [\hat{\pi}_n(z), \hat{\pi}_m(w)] \tag{4.3}$$

$$[\hat{\phi}_n(z), \hat{\pi}_m(w)] = i\hbar\delta_{nm}\delta^{(3)}(\vec{z} - \vec{w}) \tag{4.4}$$

However, just like Dirac's rule this procedure does not always produce a consistent quantum theory and in the case of quantising the Dirac Lagrangian we require the analogous anti-commutation relations because of the spin-statistics theorem [17].

4.2 Fock Space

After second quantisation in general we obtain an expression for the Hamiltonian operator in terms of various creation and annihilation operators (e.g. $a^{r\dagger}_{\vec{p}_1}, b^{s\dagger}_{\vec{p}_2}, \ldots$ and $a^{r}_{\vec{p}_1}, b^{s}_{\vec{p}_2}, \ldots$ respectively). Subsequently we define the vacuum state to be annihilated by all particle annihilation operators

$$a^r_{\vec{p}_1}|0\rangle = b^r_{\vec{p}_1}|0\rangle = \cdots = 0 \tag{4.5}$$

The indistinguishable n-particle state is given by

$$a^{r_1\dagger}_{\vec{p}_1} a^{r_2\dagger}_{\vec{p}_2} \cdots a^{r_n\dagger}_{\vec{p}_n}|0\rangle = |r_1, \vec{p}_1; r_2, \vec{p}_2; \ldots; r_n, \vec{p}_n\rangle \in H^{\otimes n} \tag{4.6}$$

$$H^{\otimes n} \coloneqq \bigotimes_{k=1}^{n} H \tag{4.7}$$

Where $H^{\otimes n}$ is the n-particle Hilbert space ($H^0 = \mathbb{C}$ corresponds to the vacuum), and an arbitrary quantum state is an element of the Fock space

$$F(H) \coloneqq \bigoplus_{n=0}^{\infty} H^{\otimes n} \tag{4.8}$$

Thus the probability amplitude of a state transitioning from one particle to another is

$$\langle r_2, \vec{p}_2 | r_1, \vec{p}_1\rangle = \left\langle 0 \,\middle|\, a^{r_2}_{\vec{p}_2} a^{r_1\dagger}_{\vec{p}_1} \,\middle|\, 0 \right\rangle \coloneqq \left\langle a^{r_2}_{\vec{p}_2} a^{r_1\dagger}_{\vec{p}_1} \right\rangle \tag{4.9}$$

which generalises to transitions between arbitrary quantum states. Hence in quantum field theory the study of probabilities is characterised by vacuum expectation values of operators and their compositions.

5 Path Integral Formulation of Quantum Mechanics

5.1 The Propagator

In quantum mechanics, the propagator is the probability amplitude of observing a position x_n at time t_n, given the particle is at postion x_1 at time t_1. This can be written as the scalar product of the two quantum states,

$$K(x', t'; x, t) := \langle x't'|xt\rangle. \tag{5.1}$$

We can rewrite this propagator as an integral over all possible paths the particle can take. We call this the Feynman path integral.

First, we discretise the time between t_n and t as

$$t_{n+1} := t' > t_n > t_{n-1} > ... > t_2 > t_1 > t =: t_0,$$

giving $n+1$ equal peices $\Delta t = t_j - t_{j-1}$ for $1 \leq j \leq n+1$.

Since position eigenstates are complete, we have

$$\int dx_j |x_j t_j\rangle\langle x_j t_j| = \mathbb{I}, \tag{5.2}$$

meaning we can rewite (5.1) as

$$\begin{aligned} K(x', t'; x, t) = \int dx_n dx_{n-1}...dx_1 \langle x_{n+1}t_{n+1}|x_n t_n\rangle\langle x_n t_n|x_{n-1}t_{n-1}\rangle... \\ ...\langle x_2 t_2|x_1 t_1\rangle\langle x_1 t_1|x_0 t_0\rangle. \end{aligned} \tag{5.3}$$

This can be thought of as integrating over all possible paths, as depicted in Figure 1.

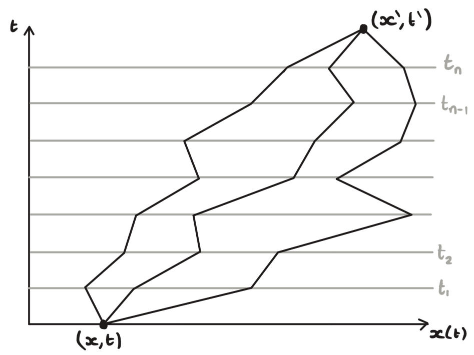

Figure 1: Three possible paths from (x, t) to (x', t') with discretised time interval.

Dirac and Feynman cite(dirac, feynman) showed that for small Δt the path of a particle is determined by the classical action, S, via the following,

$$\langle x_j t_j | x_{j-1} t_{j-1} \rangle = A\ exp\left\{\frac{i}{h} S(j, j-1)\right\} \tag{5.4}$$

where A is a normalisation constant.

S is the classical action determined by the time integral of the Lagrangian. As we let $\Delta t \longrightarrow 0$ we can approximate the path as a straight line, allowing us to calculate the Lagrangian, and hence the action.

$$\begin{aligned} S(j, j-1) &= \int_{t_{j-1}}^{t_j} dt L_{classical}(x, \dot{x}) \\ &= \int_{t_{j-1}}^{t_j} dt \left[\frac{\mu \dot{x}^2}{2} - V(x)\right] \\ &= \Delta t \left[\frac{\mu}{2}\left(\frac{x_j - x_{j-1}}{\Delta t}\right)^2 - V\left(\frac{x_j + x_{j-1}}{2}\right)\right] \end{aligned} \tag{5.5}$$

$$arethesesecondandthirdlinesneccessary?$$

Note that one can show $A = \left(\frac{\mu}{2\pi\hbar i \Delta t}\right)^{\frac{1}{2}}$.

Subbing (5.4) into (5.3) and taking the limit as $N \longrightarrow \infty$ and $\Delta t \longrightarrow 0$ gives

$$\begin{aligned} K(x', t'; x, t) &= \lim_{\substack{n\to\infty \\ \Delta t \to 0}} \left[\left(\frac{\mu}{2\pi\hbar i \Delta t}\right)^{\frac{n+1}{2}} \int dx_n dx_{n-1} ... dx_1 \prod_{j=1}^{n} exp\left\{\frac{i}{h} S(j, j-1)\right\}\right] \\ &= \lim_{\substack{n\to\infty \\ \Delta t \to 0}} \left[\left(\frac{\mu}{2\pi\hbar i \Delta t}\right)^{\frac{n+1}{2}} \int dx_n dx_{n-1} ... dx_1\ exp\left\{\frac{i}{h} \sum_{j=1}^{n} S(j, j-1)\right\}\right] \\ &= \lim_{\substack{n\to\infty \\ \Delta t \to 0}} \left[\left(\frac{\mu}{2\pi\hbar i \Delta t}\right)^{\frac{n+1}{2}} \int dx_n dx_{n-1} ... dx_1\ exp\left\{\frac{i}{h} \sum_{j=1}^{n} \int_{j-1}^{j} L_{clasical}(x, \dot{x})\right\}\right] \\ &= \lim_{\substack{n\to\infty \\ \Delta t \to 0}} \left(\frac{\mu}{2\pi\hbar i \Delta t}\right)^{\frac{n+1}{2}} \int dx_n dx_{n-1} ... dx_1\ exp\left\{\frac{i}{h} \sum_{j=1}^{n} \int_{t_0}^{n+1} L_{clasical}(x, \dot{x})\right\} \end{aligned} \tag{5.6}$$

$$needtofixthisequation \tag{5.7}$$

Proposition 5.1. *The Feynman path integral is equivalent to Schrödinger's theory.*

5.2 The Partition Function

We consider a system that evolves from a vacuum state at $t \to -\infty$ back to a vacuum state at $t \to \infty$, in the presence of a source. A vacuum state is a groundstate with zero quantum number (what?).

The source is introduced in the Lagrangian by

$$L \to L + \hbar J(t)x(t), \tag{5.8}$$

where $J(t)$ is the source and $x(t)$ is the path.

We define the partition function as the transition amplitude in the presence of this source,

$$Z[J] := \langle 0, -\infty; 0, \infty \rangle^J, \tag{5.9}$$

which one can prove is equivalent to

$$Z[J] = \int d[x(t)] exp \left\{ \frac{i}{\hbar} \int_{-\infty}^{\infty} dt(L + \hbar J(t)x(t) + \frac{1}{2} i\epsilon x^2) \right\}. \tag{5.10}$$

what is epsilon?

5.3 Time Ordered Products

AKA greens functions

5.4 Functional Derivatives of the Partition Function

This is why the partition function is aolso called generating functional

6 Generalising the Path Integral to Quantum Field Theory

In the same way we generalised classical mechanics to classical field theory in section —, we can generalise quantum mechanics to quantum field theory. That is, instead of considering trajectories of a particle x(t), we

maybe talk about why deriving the PI approach to QFT is difficult

talk about how generating functional generalises and how time ordered products generalise to operators I think?

7 Anomalies in QFT

7.1 Symmetries & Slavnov-Taylor identities

We now extend our investigation of symmetries from classical fields to a quantised theory. Suppose the classical action admits a local symmetry such that the path integral measure is invariant

$$\phi_\mu \mapsto \phi_\mu + \epsilon \vartheta_\mu(\phi, x) \tag{7.1}$$

$$\mathcal{D}\phi \mapsto \mathcal{D}\phi \, \det\left(\frac{\partial \phi'_\mu(x)}{\partial \phi_\nu(y)}\right) = \mathcal{D}\phi \tag{7.2}$$

where $K_{\mu\nu} = \frac{\partial \phi'_\mu(x)}{\partial \phi_\nu(y)}$ is the transformation Jacobian. Thus the partition function is invariant and for a source J^μ corresponding to ϕ_μ we can write [ADD PROOF L8R]

$$\int d^4x \, J^\mu(x) \langle \vartheta_\mu(\phi, x) \rangle_J = 0 \tag{7.3}$$

The quantum action is $W[J] = -i \ln Z[J]$, and we define the effective quantum action as the Legendre transform

$$\Gamma[\varphi] = W[J] - \int d^4x \, J^\mu(x) \varphi_\mu(x) \quad ; \quad \varphi_\mu := \langle \phi_\mu \rangle_J \tag{7.4}$$

Taking the functional derivative we obtain

$$J^\mu(x) = -\frac{\delta \Gamma[\varphi]}{\delta \varphi_\mu(x)} \tag{7.5}$$

Subsequently using (9.3) and the definition of a functional derivative

$$\delta \Gamma[\varphi_\mu; \langle \vartheta_\mu(\phi, x) \rangle_J] = 0 \tag{7.6}$$

Thus the effective quantum action is invariant under the transformations

$$\varphi_\mu \mapsto \varphi_\mu + \epsilon \langle \vartheta_\mu(\phi, x) \rangle_J \tag{7.7}$$

such a relation is called a Slavnov-Taylor identity. If the transformation is a linear symmetry such that

$$\vartheta_\mu = \Theta_\mu[\phi, x] = \alpha_\mu(x) + \int d^4y \, \beta^\nu_\mu(x, y) \phi_\nu(y) \tag{7.8}$$

the transformation Jacobian becomes field independent

$$K_{\mu\nu} = \frac{\delta}{\delta \phi_\nu(y)} [\phi_\mu + \epsilon \Theta_\mu(\phi, x)] = \delta_{\mu\nu} \delta^{(4)}(x - y) + \epsilon \beta^\nu_\mu(x, y) \tag{7.9}$$

so the normalised correlators are invariant. Moreover, we have $\langle \Theta_\mu(\phi, x) \rangle_J = \Theta_\mu(\varphi, x)$ which implies the effective quantum action is invariant under the same transformation as the classical action, namely,

$$\varphi_\mu \mapsto \varphi_\mu + \epsilon \Theta_\mu(\varphi, x) \tag{7.10}$$

is a symmetry. In this case the quantum theory inherits the classical symmetry and it is impossible to violate the symmetry via quantum effects granted a regularisation procedure which respects the effective quantum action's symmetry is employed. However, it is possible that no regularisation can preserve a classical symmetry, such a symmetry is called an anomaly and is characterised using the Atiyah–Singer index theorem. Returning to the Dirac Lagrangian, the vector and axial symmetries are linear in the fields but without identifying symmetry preserving regularisations their accession to a quantum theory is undetermined.

7.2 The Path Integral measure

To further investigate the effect of transformations on the partition function we must introduce a more precise notion a path integral for fermionic fields.

We decompose the Dirac spinor and its adjoint using an orthonormal basis $\{\phi_n(x)\}$ of the Dirac operator's eigenfunctions and the independent Grassmann variables θ_n and $\bar{\xi}_m$

$$\psi = \sum_n \theta_n \phi_n(x) = \sum_n \theta_n \langle x|n\rangle \tag{7.11}$$

$$\bar{\psi} = \sum_m \bar{\xi}_m \phi_m^\dagger(x) = \sum_m \bar{\xi}_m \langle m|x\rangle \tag{7.12}$$

Hence the path integral measure can be seen as a transformation of a Grassmann measure under a change of variables

$$\theta \mapsto \theta_n \langle x|n\rangle \,, \quad \bar{\xi} \mapsto \bar{\xi}_m \langle m|x\rangle \tag{7.13}$$

$$\mathcal{D}\psi\mathcal{D}\bar{\psi} = [\det\langle x|n\rangle \det\langle m|x\rangle]^{-1} \lim_{\substack{N\to\infty \\ M\to\infty}} \left\{ \prod_n^N d\theta_n \prod_m^M d\bar{\xi}_m \right\} = \lim_{N\to\infty} \prod_n^N d\theta_n d\bar{\xi}_n \tag{7.14}$$

Under the infinitesimal vector transformation the Dirac spinor becomes

$$\psi' = \sum_n \theta_n' \phi_n(x); \quad \theta_n' = e^{i\vartheta(x)}\theta_n \tag{7.15}$$

and similarly for the adjoint spinor. Thus the path integral measure transforms as

$$\mathcal{D}\psi\mathcal{D}\bar{\psi} \mapsto \left[\det\left(e^{i\vartheta(x)\mathbb{I}}\right) \det\left(e^{-i\vartheta(x)\mathbb{I}}\right)\right]^{-1} \mathcal{D}\psi\mathcal{D}\bar{\psi} \tag{7.16}$$

and using the identity [move this bcz it doesnt apply here anymore]

$$\det\left(e^A\right) = e^{\mathrm{Tr}(A)} \tag{7.17}$$

we see the measure is invariant; the vector symmetry is retained by the quantum theory and the correlators are gauge invariant. However, this is not the case for the axial transformation where the Grassmann variables transform as

$$\theta_n' = \sum_m C_{nm}\theta_m, \quad \bar{\xi}_n' = \sum_m C_{nm}\bar{\xi}_m \tag{7.18}$$

$$C_{nm} = \delta_{nm} + i\int d^4x\, \vartheta(x)\phi_n^\dagger(x)\gamma^5\phi_m(x) \tag{7.19}$$

and the path integral measure changes by the Jacobian

$$J[\vartheta] = (\det C)^{-2} = \exp\left[-2\mathrm{Tr}(\ln C)\right] \tag{7.20}$$

Expanding the matrix logarithm to first order in the infinitesimal parameter

$$\ln(\mathbb{I} + \epsilon A) = \epsilon A \tag{7.21}$$

we obtain

$$J[\vartheta] = \exp\left[-2\text{Tr}\left(i\int d^4x\,\vartheta(x)\phi_n^\dagger(x)\gamma^5\phi_m(x)\right)\right] \tag{7.22}$$

$$J[\vartheta] = \exp\left[-2i\int d^4x\,\vartheta(x)\sum_n \phi_n^\dagger(x)\gamma^5\phi_n(x)\right] \tag{7.23}$$

The sum appearing in the Jacobian is clearly divergent (by orthonormality) and requires the introduction of a regulator or ultraviolet cut-off to obtain a non-singular Jacobian. However, there is a much more elegant approach available involving the Atiyah–Singer index theorem but first we show the relation between the axial anomaly and the axial current. Henceforth, we define the axial anomaly $\mathcal{A}$ via the Jacobian of the path integral measure

$$J[\vartheta] = \exp\left[-\int d^4x\,\vartheta(x)\mathcal{A}\right] \tag{7.24}$$

under the axial transformation the partition function becomes

$$Z'[A_\mu, \vartheta] = \int \mathcal{D}\psi'\mathcal{D}\bar{\psi}'\,\exp\left[\int d^4x\,\bar{\psi}'(i\not{D} - m)\psi' - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right] \tag{7.25}$$

but this is just a relabeling of the integration variables and thus the partition function is invariant. Considering the change in the QED lagrangian we have

$$Z[A_\mu, \vartheta] = \int \mathcal{D}\psi\mathcal{D}\bar{\psi}\,J[\vartheta]\exp\left[\int d^4x\,\mathcal{L}_{\text{QED}} + \vartheta(x)(\partial_\mu j_5^\mu - 2im\bar{\psi}\gamma^5\psi)\right] \tag{7.26}$$

Thus to first order in $\vartheta(x)$ we have

$$Z[A_\mu, \vartheta] = \int \mathcal{D}\psi\mathcal{D}\bar{\psi}\,\left\{1 + \int d^4x\,\vartheta(x)(\partial_\mu j_5^\mu - 2im\bar{\psi}\gamma^5\psi - \mathcal{A})\right\}\exp\left[\int d^4x\,\mathcal{L}_{\text{QED}}\right] \tag{7.27}$$

by the fundamental lemma of the calculus of variations we attain the anomalous divergence for the axial current

$$\partial_\mu j_5^\mu = 2im\bar{\psi}\gamma^5\psi + \mathcal{A} \tag{7.28}$$

7.3 Atiyah–Singer index Theorem

[Here our task is to show (maybe just state) what the Jacobians of the vector and axial symms are and show that we can always write a relationship between the index of a differential operator and the divergence of a current. By using the AST we should find the analytical index and topological index of a differential operator coincide. My guess is the topological index should be the same using any 'sensible' regularisation thus the anomaly should persist.]

Given an eigenfunction of the Dirac operator $\not{D}$, ϕ_n, with an eigenvalue $\lambda_n \neq 0$,$\gamma^5\phi_n$ is also an eigenfunction

$$\not{D}(\gamma^5\phi_n) = -\gamma^5\not{D}(\phi_n) = -\lambda_n\gamma^5\phi_n \tag{7.29}$$

and the two are orthogonal as they have different eigenvalues. However, when the eigenvalue is zero ϕ_n and $\gamma^5\phi_n$ correspond to $\lambda_n = 0$ and are called zero modes of $\not{D}$. We also have that γ^5 is hermitian thus by the spectral theorem there is a basis that diagonalises γ^5 and spans $\mathrm{Ker}(i\not{D})$. The eigenvalues of γ^5 are ± 1 (because $(\gamma^5)^2 = I_4$) and we use the projection operators $P_\pm = \frac{1}{2}(I_4 \pm \gamma^5)$ to decompose spinors into their positive and negative eigenvalue (positive and negative chirality respectively) components. Ignoring the infinitesimal parameter for now, only the zero modes contribute to the integral in the Jacobian because of orthonormality and we have

$$\int d^4x \sum_n \phi_n^\dagger \gamma^5 \phi_n = \sum_n \int d^4x\, \phi_{n+}^{0\dagger}\phi_{n+}^0 - \sum_n \int d^4x\, \phi_{n-}^{0\dagger}\phi_{n-}^0 = n_+ - n_- \tag{7.30}$$

$$\int d^4x\, \mathcal{A} = 2i(n_+ - n_-) \tag{7.31}$$

where $n_\pm$ are the number of positive and negative chirality zero modes $\phi_{n\pm}^0$ respectively. This is precisely the analytical index of the Dirac operator projected onto the positive chirality subspace

$$\mathrm{index}(\not{D}_+) = n_+ - n_- \tag{7.32}$$

$$\not{D}_\pm := \not{D}P_\pm = \not{D}|_{\{\pm\}} \tag{7.33}$$

The Atiyah–Singer index theorem states this coincides with the Dirac operator's topological index. On a 4-dimensional compact manifold Ω without boundary and Riemannian curvature R this is [This is crucial; our calculations only work using a wick transformation, also add manifolds to integrals] ("Although the index theorem should be applied only to compact manifolds, it may, nevertheless, also be used in our Euclidean 4-space problem, since the conformal invariance of the theory and the as- sumption that the gauge fields decrease rapidly at infinity allow our problem to be mapped onto the surface of a 5-dimensional hypersphere which is compact" - R. JACKIW, C. REBBI 1977)

$$\mathrm{index}(\not{D}_+) = \int_\Omega \hat{A}(\Omega)\mathrm{ch}(F) \tag{7.34}$$

where the integrand, called the index density, only includes 4-form terms and the curvature form is given by $F = e\,dA = \frac{e}{2}F_{\mu\nu}dx^\mu \wedge dx^\nu$. The characteristic classes $\mathrm{ch}(F)$ and $\hat{A}(\Omega)$ are the Chern character of the curvature and Dirac genus of the manifold respectively

$$\mathrm{ch}(F) = \mathrm{Tr}\left(\exp\left[\frac{i}{2\pi}F\right]\right) \tag{7.35}$$

$$\hat{A}(\Omega) = \sqrt{\det\left(\frac{\frac{i}{4\pi}R}{\sinh\frac{i}{4\pi}R}\right)} \tag{7.36}$$

Now we pick the hypersphere $S^4 = \Omega$ so the Riemannian curvature is constant [put a reference to JACKIW and REBBI here] and the index becomes

$$\mathrm{index}(\not{D}_+) = \int_{S^4} \mathrm{ch}(F) = \frac{1}{2!}\left(\frac{i}{2\pi}\right)^2 \int_{S^4} \mathrm{Tr}(F^2) \tag{7.37}$$

and the integrand coincides with the Dirac operator's index density in Euclidean space. Subsequently, we identity the axial anomaly in Euclidean space as

$$\partial_\mu j_5^\mu - 2im\bar{\psi}\gamma^5\psi = \mathcal{A}[A_\mu] = \frac{-ie^2}{16\pi^2}\mathrm{Tr}(\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}) \tag{7.38}$$

7.4 Anomalies as obstructions to gauging

We now study the effect of gauging an anomalous symmetries in the case of the Dirac Lagrangian's axial symmetry. For the theory of massless Dirac fermions coupling to an axial gauge field in eq5.22, the Euclidean path integral is

$$\int \mathcal{D}\psi\mathcal{D}\bar{\psi}\,\exp\left[-\int d^4x\,\bar{\psi}i\widetilde{\not{D}}\psi - \frac{1}{4}\mathrm{Tr}(\Gamma_{\mu\nu}\Gamma^{\mu\nu})\right] \tag{7.39}$$

The classical action is gauge-invariant by construction, thus the only change in the path integral under a gauge transformation is from the Jacobian

$$\int \mathcal{D}\psi'\mathcal{D}\bar{\psi}'\,e^{-S[\psi',\bar{\psi}']} = \int \mathcal{D}\psi\mathcal{D}\bar{\psi}\,J[\vartheta]e^{-S[\psi,\bar{\psi}]} \tag{7.40}$$

where the Jacobian takes the same form as in eq9.23 for an orthonormal basis of eigenfunctions $\{\phi_n(x)\}$ of the operator $\widetilde{\not{D}}$. We now proceed via Fujikawa's method of gauge-invariant mode cut-off to calculate the Jacobian by introducing a smooth regulator $f(x)$ with all derivatives vanishing at infinity

$$J[\vartheta] = \exp\left[-2i\lim_{N\to\infty}\left\{\sum_{n=1}^{N}\int d^4x\,\vartheta(x)\phi_n^\dagger\gamma^5\phi_n\right\}\right] \tag{7.41}$$

$$J[\vartheta] = \exp\left[-2i\lim_{M\to\infty}\left\{\sum_{n=1}^{\infty}\int d^4x\,\vartheta(x)\phi_n^\dagger\gamma^5 f\left(\frac{\lambda_n^2}{M^2}\right)\phi_n\right\}\right] \tag{7.42}$$

$$J[\vartheta] = \exp\left[-2i\lim_{M\to\infty}\mathrm{Tr}\left\{\vartheta(x)\gamma^5 f\left(\frac{\widetilde{\not{D}}^2}{M^2}\right)\right\}\right] \tag{7.43}$$

where the functional trace is taken over spacetime using the basis $\{\phi_n(x)\}$. Now we compute the series using momentum space

$$\phi_n(x) = \int \frac{d^4k}{(2\pi)^2}\,\tilde{\phi}_n(k)e^{ikx} \tag{7.44}$$

$$\sum_{n=1}^{\infty}\phi_n^\dagger\gamma^5 f\left(\frac{\widetilde{\not{D}}^2}{M^2}\right)\phi_n = \int \frac{d^4k d^4p}{(2\pi)^4}\sum_{n=1}^{\infty}e^{-ipx}\tilde{\phi}_n^\dagger(p)\gamma^5 f\left(\frac{\widetilde{\not{D}}^2}{M^2}\right)\tilde{\phi}_n(k)e^{ikx} \tag{7.45}$$

$$= \int \frac{d^4k}{(2\pi)^4}\mathrm{Tr}\left\{\gamma^5 f\left(\frac{(\widetilde{D}_\mu + ik_\mu)(\widetilde{D}^\mu + ik^\mu)}{M^2} + \frac{iq[\gamma^\mu,\gamma^\nu]\Gamma_{\mu\nu}}{4M^2}\right)\right\} \tag{7.46}$$

where we have used the identities

$$\sum_{n=1}^{\infty} \tilde{\phi}_n^\dagger(p) A \tilde{\phi}_n(k) = \delta(k-p)\mathrm{Tr}(A) \tag{7.47}$$

$$\widetilde{\not{D}}^2 = \widetilde{D}_\mu \widetilde{D}^\mu + \frac{iq}{4}[\gamma^\mu, \gamma^\nu]\Gamma_{\mu\nu} \tag{7.48}$$

$$e^{-ikx} f(\partial_\mu) e^{ikx} = f(\partial_\mu + ik_\mu) \tag{7.49}$$

Now we rescale the momenta via $k_\mu \to M k_\mu$ and the series becomes

$$M^4 \int \frac{d^4k}{(2\pi)^4} \mathrm{Tr}\left\{\gamma^5 f\left(\frac{\widetilde{D}_\mu \widetilde{D}^\mu + 2iMk_\mu \widetilde{D}^\mu}{M^2} - k^2 + \frac{iq[\gamma^\mu, \gamma^\nu]\Gamma_{\mu\nu}}{4M^2}\right)\right\} \tag{7.50}$$

expanding the Taylor series of $f(x)$ about $-k^2 = -k_\mu k^\mu$, and using the trace properties of the gamma matrices the only term non-vanishing in the limit $M \to \infty$ is the term quadratic in $\Gamma_{\mu\nu}$

$$\frac{M^4 q^2}{2! M^4} \mathrm{Tr}(\epsilon^{\mu\nu\rho\sigma}\Gamma_{\mu\nu}\Gamma_{\rho\sigma}) \int \frac{d^4k}{(2\pi)^4} f''(-k^2) \tag{7.51}$$

and we have

$$J[\vartheta] = \exp\left[\int d^4x \, \frac{-iq^2}{16\pi^2} \mathrm{Tr}(\epsilon^{\mu\nu\rho\sigma}\Gamma_{\mu\nu}\Gamma_{\rho\sigma})\vartheta(x)\right] \tag{7.52}$$

thus there is gauge-variance in the path integral measure and the quantum theory is inconsistent. In the general case, gauging an anomalous symmetry will produce an inconsistent quantum theory as constructing a gauge invariant classical action ensures the shift in the measure makes the path integral change under a gauge transformation.

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